$$\frac{1}{m}(Y - X\theta)^T(Y - X\theta) + \lambda \theta^T\theta$$

$$= \frac{1}{m}\left(\theta^T X^T X\theta - 2Y^T X\theta + Y^T Y\right) + \lambda\theta^T\theta$$

$$= \frac{2}{m}\theta^T X^T X - \frac{2}{m}Y^T X + \lambda 2\theta^T = 0$$

$$\Rightarrow \quad \theta^T X^T X - Y^T X + m\lambda\theta^T = 0$$

$$\Rightarrow Y^T X = \theta^T X^T X + m\lambda\theta^T$$

$\lambda' = m\lambda \Rightarrow Y^T X = \theta^T X^T X + \lambda\theta^T$

$$\Rightarrow Y^T X = \theta^T\left(X^T X + \lambda I\right)$$

$$\left(X^T X + \lambda'\right)^{-1} Y^T X = \theta^T$$

$$\left(X^T X + \lambda' I\right) X^T Y = \theta$$

Ridge regression estimator.